

ERNEST WONG

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EDUCATION

Simon Fraser University - B.Sc. Computer Science

University of Glasgow - Field Studies in Cybercrime

British Columbia Institute of Technology - Diploma, Cloud Computing Systems, with Distinction

Expected 2027

2024

2021

EXPERIENCE

ISED | Government of Canada

JAN 2025 - PRESENT

Software Engineer Intern | Java, Python, GraphQL, Jenkins, AWS, Salesforce

- Led production deployments for a **\$100M+ patent submission platform**, coordinating AWS Cloud Ops, Salesforce, Java teams, database changes, sanity testing, and release verification.
- Built Java/GraphQL drift-detection tooling across **8+ environments** and automated daily Jenkins comparisons against known-good systems to isolate configuration mismatches during production investigations.
- Developed Python/Selenium automation for CI/CD setup across **50+ applications**, standardizing Bitbucket, Bamboo, Nexus, and IBM WebSphere configuration.
- Resolved Salesforce Lightning UI and API defects affecting bilingual submissions and high-volume online patent filing workflows.

Elections Canada | Government of Canada

MAY 2026 - PRESENT

Data Analyst Co-op | Python, SQL, Denodo, Power BI, Tableau

- Engineered a Python extraction pipeline that converts recurring Tableau-generated PDF dashboards into normalized, **Power BI-ready CSV datasets**, replacing manual transcription and recovering previously missed metrics.
- Translated complex Power BI transformation logic into reusable **Denodo VQL and SQL views**, centralizing reporting rules for downstream analysis.
- Implemented cross-system validation across Denodo, People Insights, and source systems to flag missing and inconsistent records before reporting.

SFU Satellite Design Team | UBC Orbit Collaboration

SEP 2023 - PRESENT

Mission Operations Developer | Embedded Systems Lead | Python, React, AWS, OpenShift, Raspberry Pi

- Developed the **ALEASAT spacecraft dashboard** with React, AWS, and OpenShift APIs for telemetry and command workflows supporting a European Space Agency mission targeting a 2028 SpaceX launch.
- Led PDR, CDR, and FRR for a high-altitude balloon mission reaching **~30 km**, integrating Python flight software with **10+ sensors and subsystems** on a Raspberry Pi.
- Migrated outreach websites from Squarespace to React and GitHub Pages, reducing annual hosting costs from **\$200 to \$0**.

Simon Fraser University | Faculty of Applied Sciences

MAY 2024 - AUG 2024

Research Assistant, AI Cybersecurity | Python, Machine Learning, AI Security, Tensor Trust

- Developed a Python ML tool to remove personal information from AI models, achieving an **F1 score above 80%**, and evaluated privacy vulnerabilities using Tensor Trust.

PROJECTS

CleanListen - Data-Centric PDF Cleaning for Screen Readers

2025 - PRESENT

Independent Project | Python, scikit-learn, Pandas, PDF Processing

- Developed a line-level KEEP/SKIP classifier and labeling pipeline for noisy academic PDFs, reaching **91.2% accuracy and 89.7% KEEP F1** across 1,929 evaluated lines while filtering navigation noise for accessible listening.

LEADERSHIP & RECOGNITION

- **Leadership:** Led recurring ISED AI Coffee Chats for 300+ attendees, and presented to 100+ prospective students as an SFU Peer Tutor, automating 200 personalized mentoring emails.
- **Recognition:** CS Undergraduate Society Award (1 of 5) | AI Hackathon Finalist (Best Use of Google Gemini) | Mayor of Richmond Recognition Award

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, SQL, Bash

Frameworks & Data: React, Node.js, GraphQL, REST, Pandas, Spark, Power BI, Denodo, Salesforce

Cloud & DevOps: AWS, Docker, Kubernetes, OpenShift, Jenkins, Bamboo, Bitbucket, IBM WebSphere

Testing & Tools: Selenium, Pytest, JUnit, Cypress, Git, Jira, Postman, Maven